

Performative Prediction in a Single-Asset Market

Self-fulfilment and signal erosion under forecast adoption

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0.07 → 0.19

what a live forecaster sees (rises)
realised-return R^2

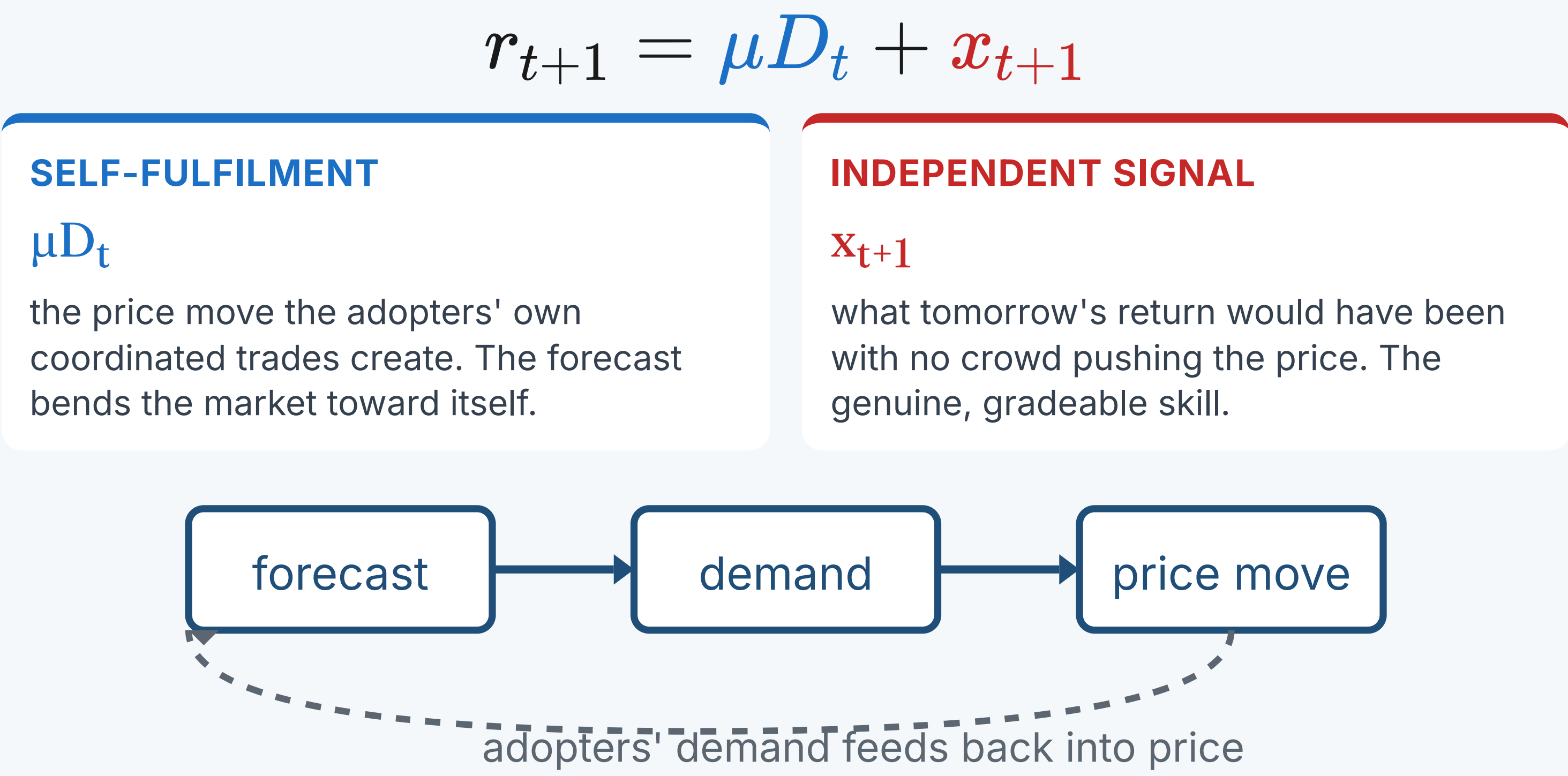
0.08 → 0.04

the genuinely independent signal (falls)
demand-adjusted R^2

0.28 → 0.45

the market's memory stiffens
effective persistence ϕ

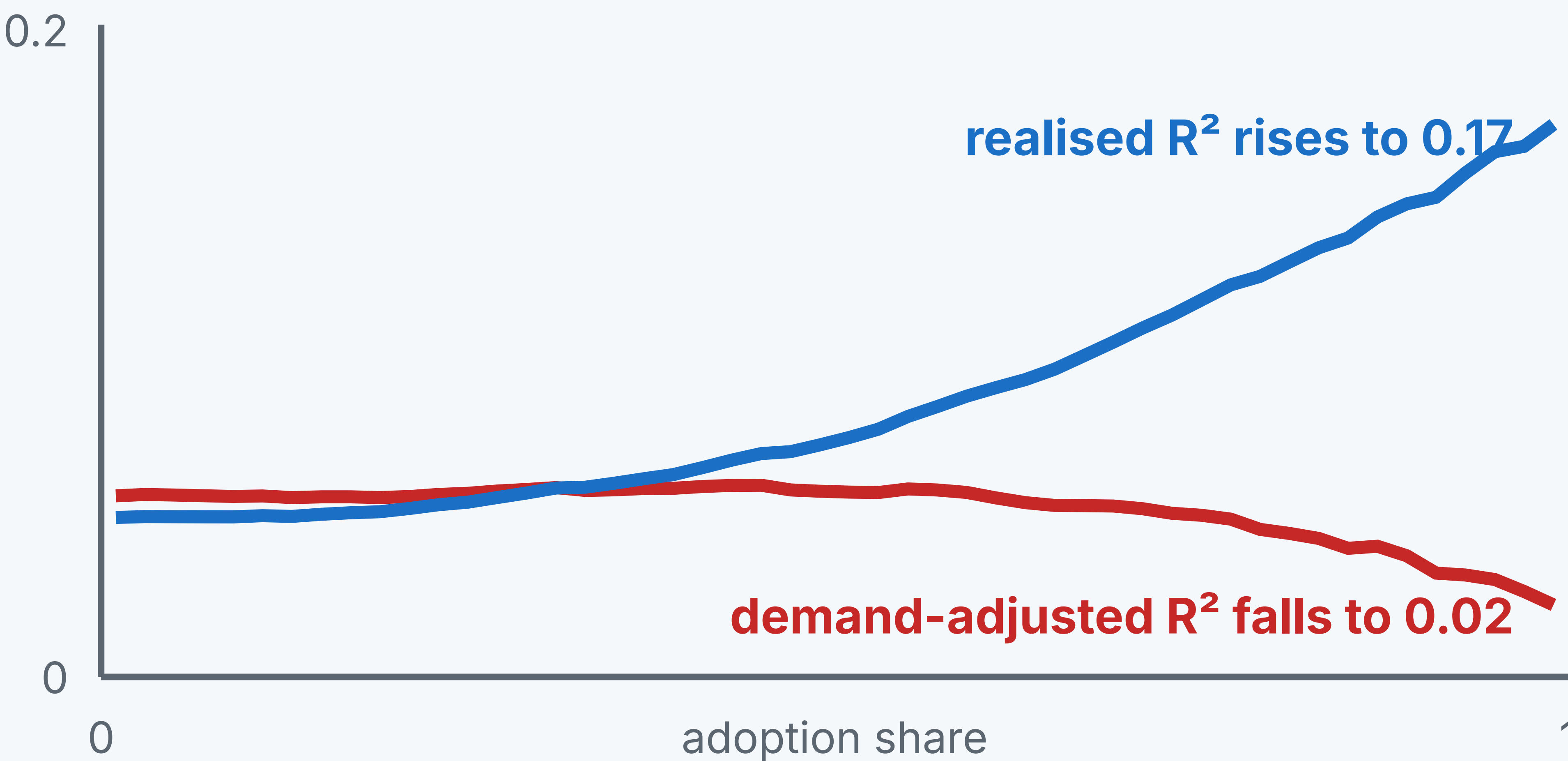
One forecast, two channels



One simulated market



As adoption rises, one forecast earns two verdicts



rolling out-of-sample R^2 against each target, extended saturated regime ($T = 30,000$). The two readings of the same forecast pull apart as the crowd grows.

The question

If everyone trades on the same forecast, does it get better, or just bend the market into agreeing? Publish one forecast of tomorrow's return and traders begin acting on it. Their coordinated buying and selling pushes the very price the forecast will later be graded against.

The model

Strip a market to its essentials: traders who buy more when they expect a higher return and hold back when risk is high, and a market maker who moves the quote to absorb their net demand (Brock and Hommes 1998; Beja and Goldman 1980; Farmer and Joshi 2002). It is the smallest market in which a shared forecast can move

Intra-period timing

1. observe returns through t , form forecasts
2. adopters and non-adopters submit orders
3. maker absorbs demand, moves the quote
4. news shock and persistence term realise

Two evaluation targets

We score the same forecast two ways on the exact same price path. The realised-return score asks the deployment question: did the price actually move as predicted? The demand-adjusted score first strips out the adopters' own price impact (the μD_t term), asking the honest question: was the signal genuinely there before the crowd pushed? One is what a live forecaster measures, the other is the real underlying skill

The dual-channel result

Here is the twist. As more traders adopt the forecast, its realised-return score climbs from 0.07 to 0.19, looking sharper and sharper. But that rise is mostly self-fulfilment: the crowd's coordinated trades drag the price toward the prediction, so the forecast is partly grading its own homework. Strip that push out and the demand-adjusted score falls the other way, from 0.08 to 0.04, meaning the genuine, independent signal is actually eroding. The same trades stiffen the market's memory too (effective persistence rises from 0.28 to 0.45), feeding the self-fulfilling loop even harder.